PROJECT REPORT

# Multiclass Sleep Health Risk Classification & Explainable AI System

An End-to-End Machine Learning Pipeline for Predicting Insomnia and Sleep Apnea from Lifestyle and Physiological Health Data

*Supervised Learning • Class-Imbalance Handling • Explainable AI (XAI)*

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| Final Model | Test Accuracy | Test Macro F1 | Dataset Size |
|---|---|---|---|
| XGBoost (Tuned) | 93.50% | 88.50% | 10,000 records |

*Dataset: Sleep Health & Lifestyle Records (10,000 individuals, 13 features) • 3 Target Classes: No Sleep Disorder · Insomnia · Sleep Apnea*
*Leakage-Free Pipeline • Stratified Cross-Validation • SHAP-Based Model Interpretability • Flask Deployment*

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# 1. Executive Summary

This report documents an end-to-end machine learning project that classifies an individual's sleep health status using demographic, lifestyle, and physiological data. The system distinguishes between three outcomes — **No Sleep Disorder**, **Insomnia**, and **Sleep Apnea** — from a dataset of 10,000 patient records spanning occupation, activity, stress, sleep quality, and cardiovascular indicators.

The project follows a leakage-aware, eight-phase pipeline: data preparation, data understanding, exploratory analysis, feature engineering, preprocessing, model development, hyperparameter tuning, and final evaluation with explainability. Five classification algorithms — Logistic Regression, Decision Tree, Random Forest, Support Vector Machine, and XGBoost — were each benchmarked under two class-imbalance strategies (class weighting and SMOTE) using Stratified 5-Fold Cross-Validation, since the target is imbalanced at roughly 78% / 12% / 10% across its three classes.

> **Headline Result**
>
> A tuned **XGBoost** classifier was the best-performing model, achieving **93.50% accuracy** and a **Macro F1-score of 88.50%** on a completely unseen, held-out test set — outperforming Logistic Regression, Decision Tree, Random Forest, and SVM under both imbalance-handling strategies tested.

Beyond raw predictive accuracy, the project prioritised clinical interpretability. SHAP (SHapley Additive exPlanations) was applied to the final XGBoost model to quantify which features drive each prediction, confirming that **Heart Rate**, **Sleep Duration**, and the engineered **Sleep Quality Index** are the dominant physiological and behavioural signals, while demographic and occupational attributes contribute comparatively little.

## 1.1 What This Project Demonstrates

- **End-to-end ML pipeline design** — from raw Excel ingestion to a deployable, explainable classifier.
- **Imbalanced-classification discipline** — stratified splitting, class weighting, and SMOTE, compared side-by-side rather than assumed.
- **Leakage-safe engineering** — preprocessing and resampling refit independently inside every cross-validation fold.
- **Feature engineering for health data** — pulse pressure, a composite sleep-quality index, and a stress–sleep interaction term, each derived without touching the target.
- **Breadth across model families** — linear, tree-based, kernel-based, and gradient-boosted classifiers.
- **Explainable AI (XAI)** — global and interaction-level SHAP analysis, turning a 'black-box' gradient-boosted model into a transparent, stakeholder-readable decision system.

# 2. Project Overview

## 2.1 Project Objective

Develop an explainable multiclass classification system that predicts an individual's sleep-disorder status — No Sleep Disorder, Insomnia, or Sleep Apnea — from demographic, occupational, lifestyle, and physiological attributes, while benchmarking multiple algorithms and imbalance-handling strategies to identify the most reliable, generalisable model.

### 2.2 Business Objective

- Flag individuals at risk of a sleep disorder from routinely available health and lifestyle data.
- Benchmark class-imbalance strategies and select the strongest model on class-aware metrics, not raw accuracy alone.
- Explain predictions with SHAP to support clinical trust and stakeholder transparency.
- Package the final model as a deployable Flask web application for real-time inference.

### 2.3 Problem Statement

Sleep disorders such as Insomnia and Sleep Apnea are frequently under-diagnosed despite measurable ties to lifestyle and physiological signals — stress, activity level, blood pressure, and heart rate. Given an individual's demographic, lifestyle, and physiological attributes, this project predicts sleep-disorder status as one of three classes, using a dataset that is naturally imbalanced (78% / 12% / 10%), which makes naïve accuracy an unreliable evaluation metric and motivates the model-selection strategy used throughout this report.

### 2.4 Dataset Description

The dataset consists of synthetically generated patient-level sleep, lifestyle, and cardiovascular records.

| Attribute | Value |
|---|---|
| Raw Records | 10,050 rows (50 exact duplicates removed) |
| Modelling Records | 10,000 records × 13 features |
| Target Variable | Sleep Disorder (No Sleep Disorder · Insomnia · Sleep Apnea) |
| Feature Groups | Demographic, Occupational, Lifestyle, Physiological, Cardiovascular |
| Missing Values | 12 of 13 features affected (up to ~5.4% per feature) |
| Duplicate Records | 50 identified and removed |
| Source | Sleep Health and Lifestyle Dataset (Kaggle, synthetically generated) |

**Data cleaning note:** the combined *Blood Pressure* field (e.g. "126/83") was split into separate *Systolic BP* and *Diastolic BP* columns to enable numerical analysis. The literal string *"None"* in the target column was explicitly preserved as a valid category during ingestion (rather than parsed as a missing value) and relabelled to *"No Sleep Disorder"*, and inconsistent categorical entries (e.g. "FEMALE", "female", "Female ") were standardised prior to encoding.

## 3. Data Preparation & Understanding

Before analysis, the raw dataset was cleaned and validated to ensure every downstream statistic and model is computed on a trustworthy, consistent table.

### 3.1 Initial Data Preparation

- **Identifier removal:** non-predictive identifier columns (e.g. Person ID) were dropped before analysis.
- **Duplicate handling:** 50 exact duplicate records were identified and removed, leaving 10,000 unique rows.
- **Categorical standardisation:** inconsistent casing and stray whitespace in Gender, Occupation, and BMI Category were normalised to a single canonical spelling per category.

- **Missing-value standardisation:** blank strings were explicitly converted to NaN so that numerical and categorical imputers could detect them consistently.
- **Blood Pressure decomposition:** the combined "Systolic/Diastolic" text field was split into two numeric columns, Systolic BP and Diastolic BP.

### 3.2 Data Quality & Validation Checks

Following the initial preparation steps, the dataset was re-validated to confirm the cleaning stage produced a leakage-free, analysis-ready table.

| Check | Result |
|---|---|
| Duplicate records remaining | 0 |
| Features with missing values | 12 of 13 (imputed later inside the pipeline, not dropped) |
| Maximum missing-value rate | ~5.4% (single feature) |
| Target class integrity | "None" correctly preserved and relabelled — not read as null |
| Categorical consistency | Standardised (no duplicate-meaning categories remain) |

### 3.3 Descriptive & Target Distribution Summary

Descriptive statistics confirmed plausible physiological ranges across the dataset (e.g. Heart Rate roughly 50–125 bpm, Sleep Duration roughly 4.5–9.5 hours), with no impossible or out-of-range values detected. The target variable, however, is meaningfully imbalanced — a fact that shapes the modelling strategy for the remainder of this report.

| Class | Count | % of Dataset |
|---|---|---|
| No Sleep Disorder | 7,813 | 78.13% |
| Insomnia | 1,237 | 12.37% |
| Sleep Apnea | 950 | 9.50% |

**Key takeaway:** a model that always predicts "No Sleep Disorder" would already score ~78% accuracy while never identifying a single at-risk patient. This motivated **Macro F1** as the primary model-selection metric, alongside class weighting and SMOTE as imbalance-handling strategies.

## 4. Exploratory Data Analysis (EDA)

Before modelling, the cleaned dataset was explored visually and statistically to understand the shape of the target, feature distributions, categorical breakdowns, correlations, and outliers.

### 4.1 Target Variable Distribution

The target variable shows a clear three-way class imbalance: healthy individuals dominate the dataset, while Insomnia and Sleep Apnea together make up roughly one in five records.

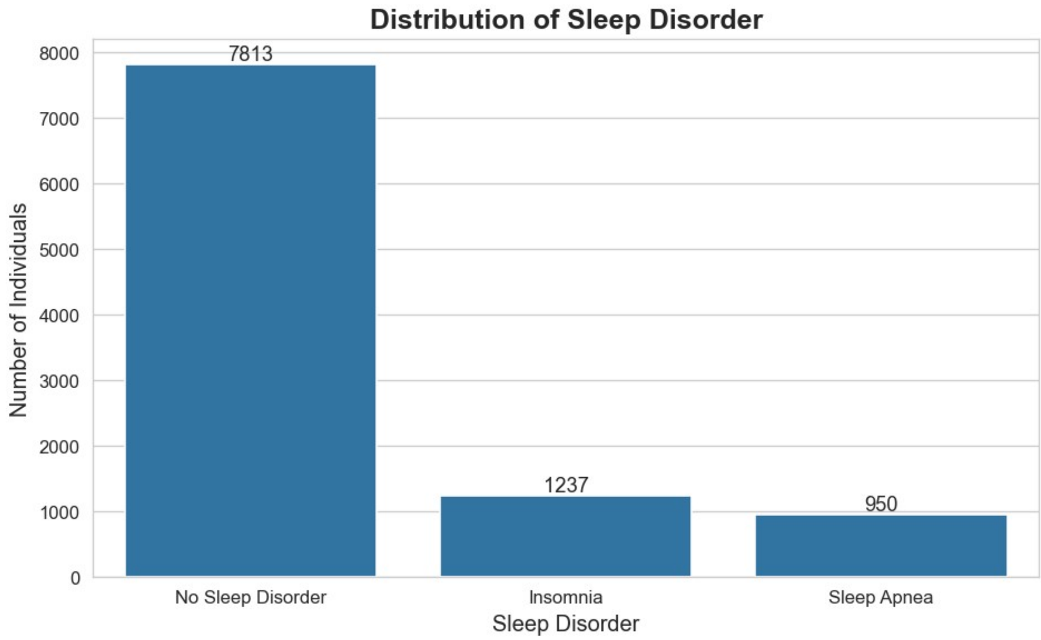


*Figure 4.1 — Distribution of the Sleep Disorder Target Variable*

**Observation:** the 78% / 12% / 10% split confirms the dataset is imbalanced rather than merely skewed, reinforcing the decision to evaluate models on Macro F1 and Balanced Accuracy rather than accuracy alone.

## 4.2 Numerical Feature Distribution

Histograms with kernel-density overlays were plotted for all nine numerical features to inspect shape, skew, and multimodality ahead of modelling.

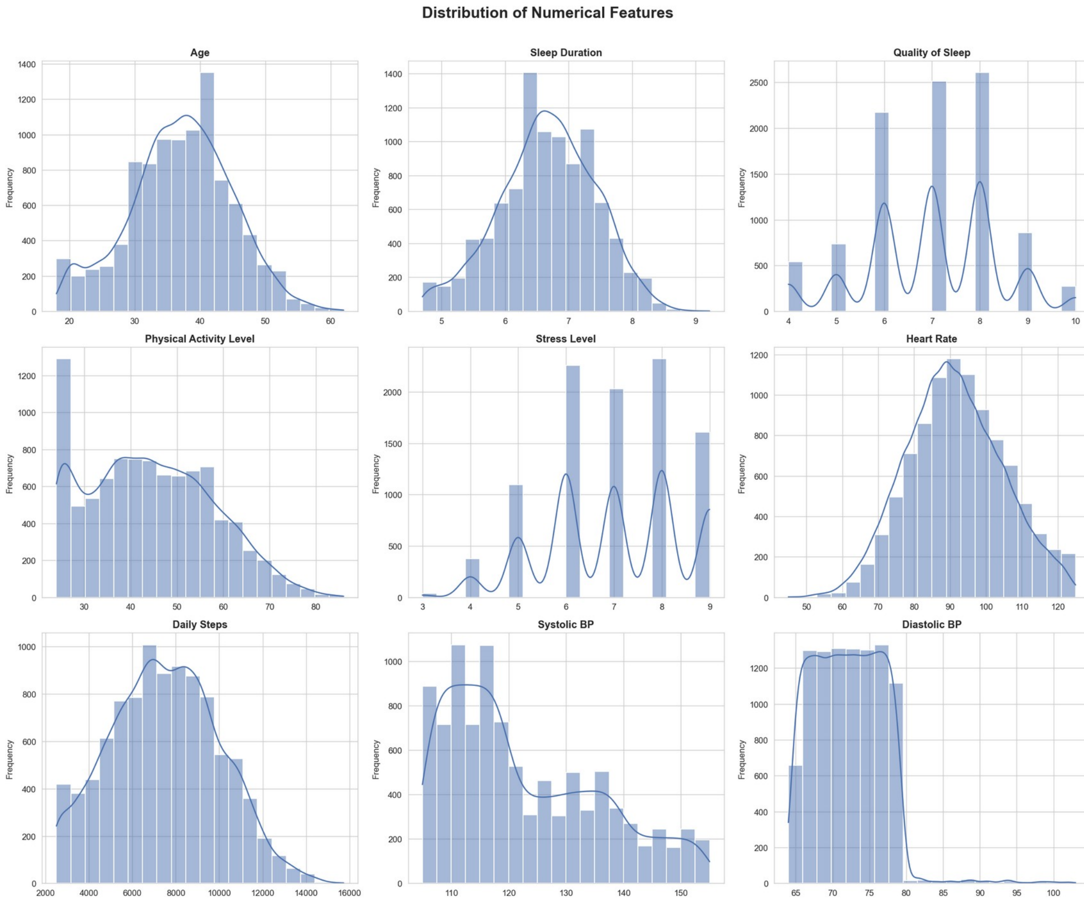


*Figure 4.2 — Distribution of Numerical Features*

**Insights:** Age and Sleep Duration are broadly bell-shaped; Stress Level and Quality of Sleep are discrete, multi-modal ordinal scales rather than continuous measures; Daily Steps and Systolic BP show mild right-skew consistent with a small subset of more sedentary, higher-risk individuals.

## 4.3 Categorical Feature Distribution

Gender, Occupation, and BMI Category were profiled to understand demographic composition and category cardinality ahead of one-hot encoding.

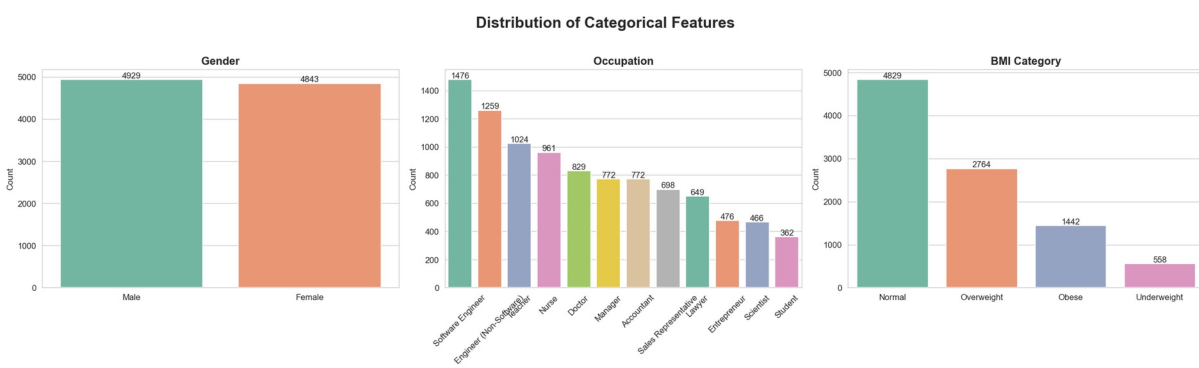


*Figure 4.3 — Distribution of Categorical Features*

**Insights:** Gender is close to evenly split (4,929 vs. 4,843). Occupation is dominated by Software Engineers, Teachers, and Nurses, with several low-frequency categories (e.g. Scientist, Student) that require careful handling to avoid sparse one-hot columns. Just under half of individuals fall in the Normal BMI category, with Overweight the next-largest group.

## 4.4 Feature Relationships with the Target

Each numerical feature was compared across the three target classes using box plots to identify which signals separate healthy individuals from those with a disorder.

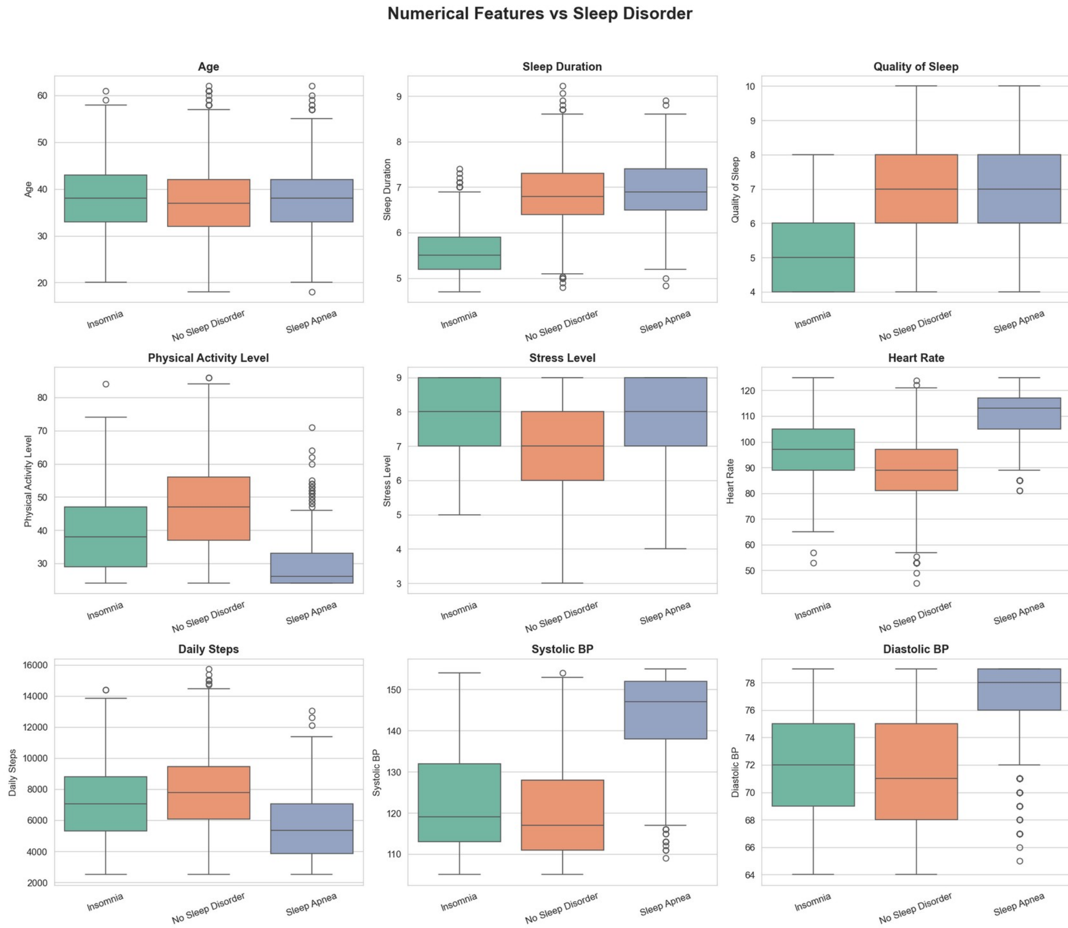


*Figure 4.4a — Numerical Features vs. Sleep Disorder*

**Insights: Sleep Apnea** patients show markedly higher Systolic BP, Diastolic BP, and Heart Rate than the other two classes, alongside lower Physical Activity Level and Daily Steps. **Insomnia** is strongly associated with shorter Sleep Duration, lower Quality of Sleep, and elevated Stress Level. Age shows only a weak relationship with disorder status.

The same comparison was repeated for the three categorical features to test for demographic or occupational association with the target.

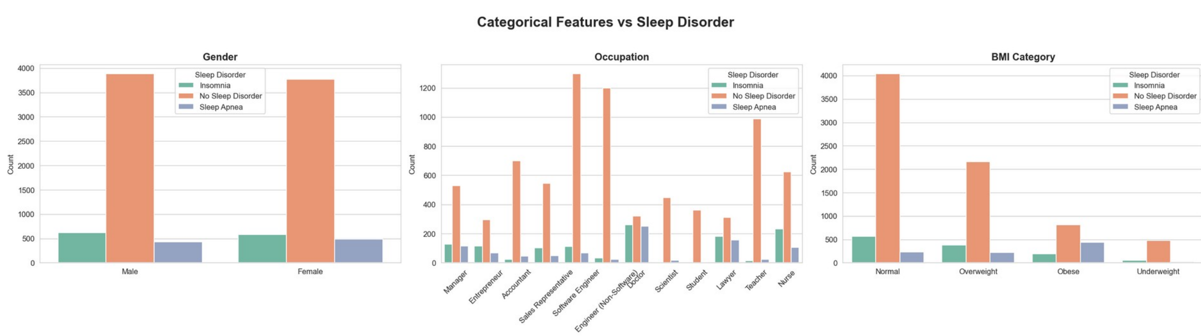


*Figure 4.4b — Categorical Features vs. Sleep Disorder*

**Insights: BMI Category** shows the clearest categorical association — Obese individuals skew heavily toward Sleep Apnea relative to Normal-weight individuals. Certain **Occupations** (Nurse, Doctor, Sales Representative) show elevated disorder rates, plausibly as a proxy for stress and shift-work patterns, while **Gender** shows minimal separation between classes.

## 4.5 Correlation Analysis

A Pearson correlation matrix of the numerical features was computed to surface linear relationships and flag potential multicollinearity ahead of model training.

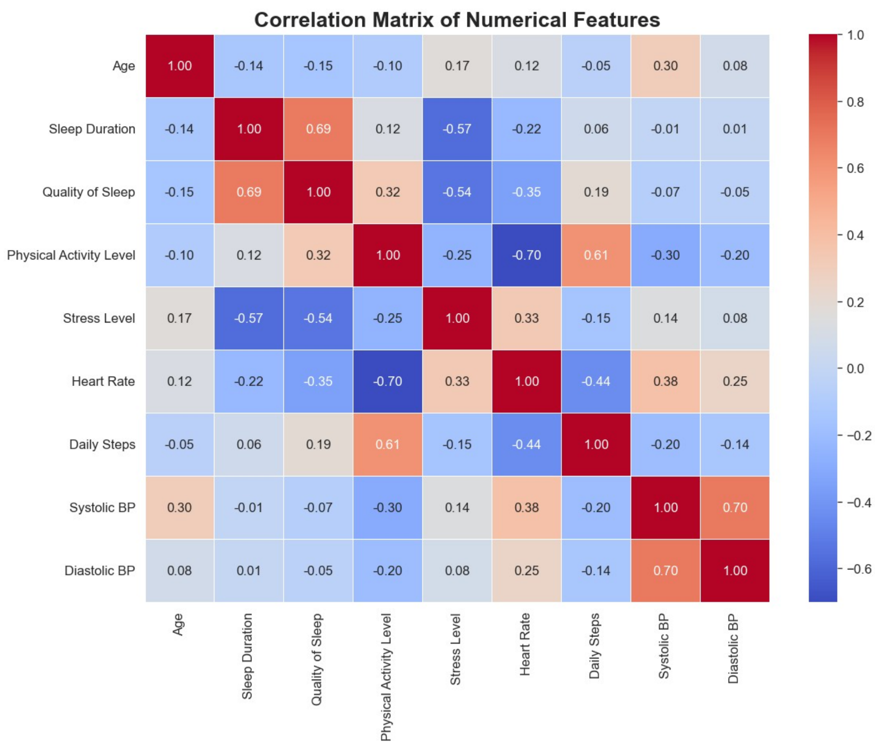


*Figure 4.5 — Correlation Matrix of Numerical Features*

**Insights:** Sleep Duration and Quality of Sleep are strongly positively correlated (r = 0.69), as expected for two related sleep metrics. Physical Activity Level and Heart Rate are strongly negatively correlated (r = −0.70), and Systolic BP / Diastolic BP are strongly positively correlated (r = 0.70), confirming these are related but non-redundant cardiovascular signals. No pair reaches the near-perfect correlation that would force feature removal, so all numerical predictors were retained for modelling.

## 4.6 Multivariate Relationship Analysis

A pair-plot of the most target-relevant numerical features (Sleep Duration, Quality of Sleep, Stress Level, Heart Rate) was generated, coloured by class, to visually inspect how well these signals separate the three sleep-disorder categories in combination.

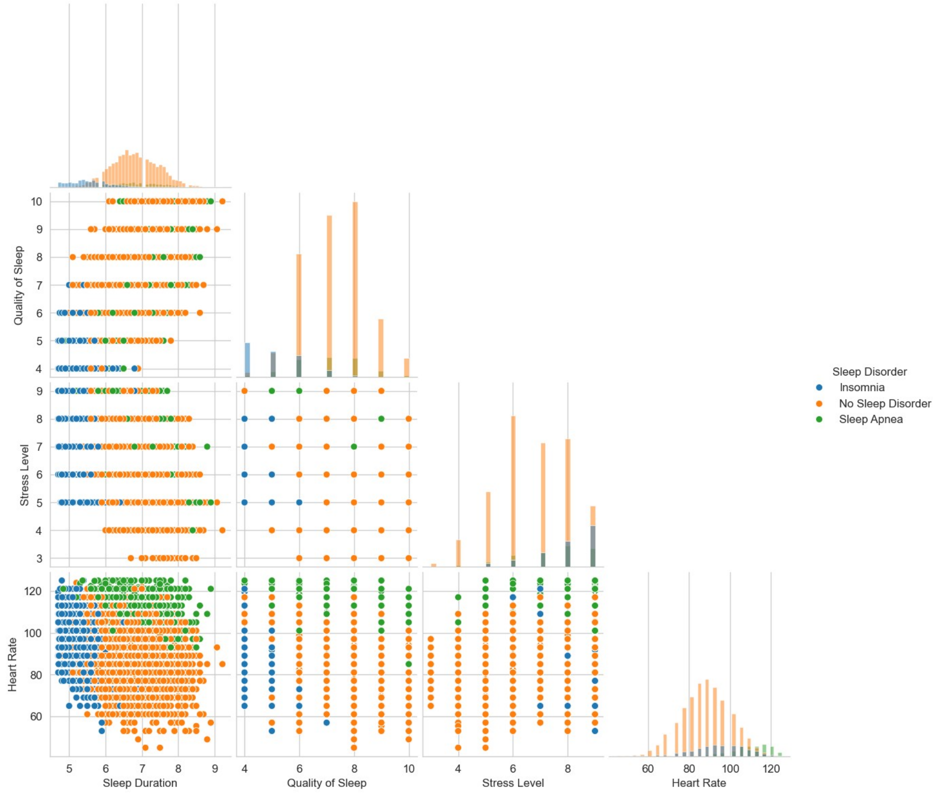


*Figure 4.6 — Pair Plot of Key Features by Sleep Disorder Class*

**Insights:** Insomnia (blue) and Sleep Apnea (green) each occupy visually distinct regions of the Sleep Duration × Heart Rate and Quality of Sleep × Heart Rate planes, while No Sleep Disorder (orange) dominates the moderate-stress, moderate-heart-rate region — evidence that even simple, linearly-separable structure exists in the feature space ahead of formal modelling.

## 4.7 Outlier Analysis

Outliers in the nine numerical features were assessed using the Interquartile Range (IQR) method and visualised with box plots.

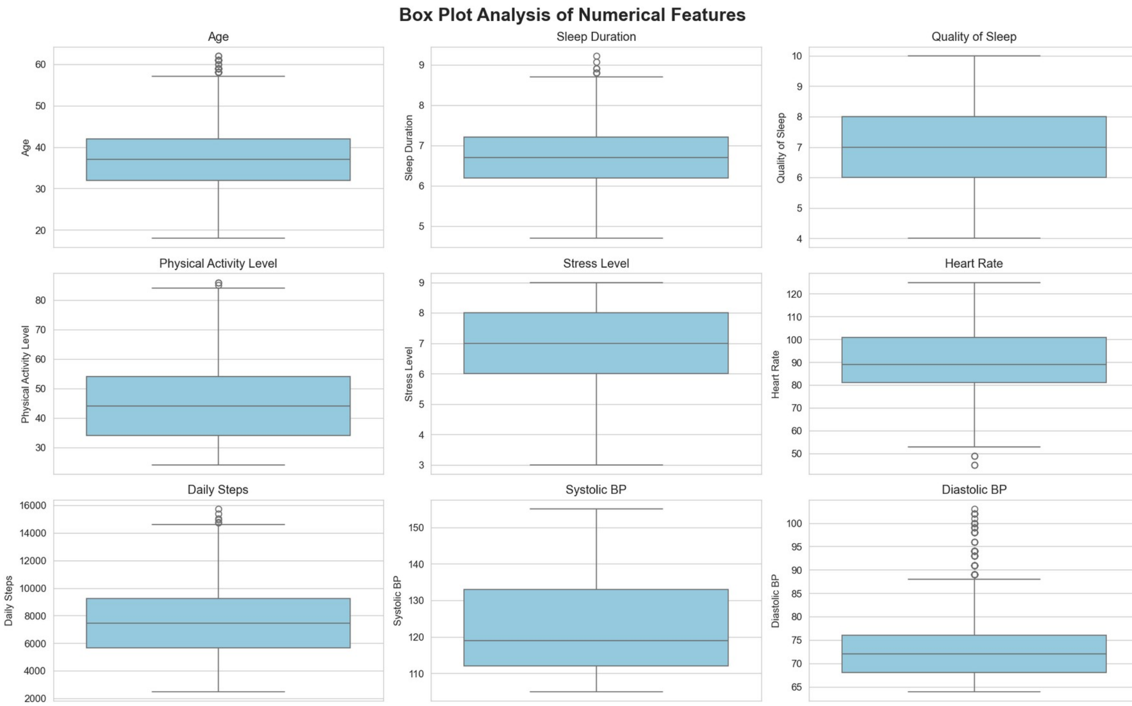


*Figure 4.7 — Box Plot Analysis of Numerical Features (Outlier Detection)*

**Key Insights:** Diastolic BP shows the highest proportion of statistical outliers, followed by Sleep Duration and Age, each still representing under 1% of observations. These points correspond to physiologically plausible individuals (e.g. very low resting heart rate, unusually high blood pressure) rather than data-entry errors, so they were retained rather than removed — consistent with the objective of modelling real-world variation, including at-risk edge cases.

> **EDA Summary**
>
> Target is imbalanced (78% / 12% / 10%) → Macro F1 and class weighting/SMOTE required.
>
> Sleep Apnea is driven by elevated blood pressure and heart rate; Insomnia by lower sleep quality, shorter duration, and higher stress.
>
> BMI Category (Obese) is the strongest categorical signal; Gender carries little predictive value.
>
> No feature pair is collinear enough to warrant removal → all predictors retained.
>
> Outliers are limited to genuine physiological variation (<1% of rows) and were kept, not removed.

# 5. Feature Engineering

Three domain-informed features were derived from existing predictors to give the models additional, clinically meaningful signal without introducing any dependency on the target variable.

## 5.1 Engineered Features

| Feature | Formula | Rationale |
|---|---|---|
| Pulse Pressure | Systolic BP − Diastolic BP | Additional cardiovascular signal beyond raw BP |
| Sleep Quality Index | ((Sleep Duration / 24) + (Quality of Sleep | Consolidated, standardised sleep-profile |

| Feature | Formula | Rationale |
|---|---|---|
| | / 10)) / 2 | score |
| Stress-Sleep Interaction | Stress Level × Sleep Duration | Captures the combined effect of stress and sleep duration |

### 5.2 Engineered Feature Validation

Each engineered feature was checked for missing values (none were introduced) and validated by comparing its group-wise mean across the three target classes.

- **Pulse Pressure** shows strong separation for Sleep Apnea — mean 66.5 vs. approximately 49 for the other two classes — directly reflecting the elevated blood pressure observed during EDA.
- **Sleep Quality Index** is lowest for Insomnia (0.368) compared to approximately 0.50 for the other classes, consolidating two related sleep metrics into a single, standardised signal.
- **Stress-Sleep Interaction** captures individuals experiencing the combined burden of high stress and short sleep duration, a pattern that raw stress or raw sleep-duration alone does not fully express.

All three engineered features separate the target classes meaningfully and were retained for modelling.

## 6. Data Preprocessing & Train-Test Split

### 6.1 Preprocessing Pipeline

All preprocessing was implemented as a single **scikit-learn ColumnTransformer**, fit exclusively on the training fold in every cross-validation split to prevent information leakage from the test set.

| Feature Type | Pipeline |
|---|---|
| Numerical | Median imputation → StandardScaler |
| Categorical | Mode imputation → OneHotEncoder |

Combining the numerical and one-hot-encoded categorical outputs expands the processed feature matrix from 13 raw columns to **30 model-ready features**.

### 6.2 Stratified Train-Test Split

Given the 78% / 12% / 10% class imbalance, a random split risks under-representing Insomnia or Sleep Apnea in either partition. A **stratified 80/20 split** was therefore used to preserve the original class proportions in both the training and test sets.

| Split | Samples | Class Proportions Preserved |
|---|---|---|
| Training Set | 8,000 | Yes (stratified) |
| Test Set | 2,000 | Yes (stratified) |

The test set remained completely untouched until the final evaluation stage in Section 8.

# 7. Model Development & Selection

## 7.1 Modeling Approach

Five classification algorithms — Logistic Regression, Decision Tree, Random Forest, Support Vector Machine, and XGBoost — were each benchmarked under two imbalance-handling strategies, **class weighting** and **SMOTE** (Synthetic Minority Over-sampling), using **Stratified 5-Fold Cross-Validation**. Preprocessing and SMOTE (where applicable) were refit independently inside every fold to guarantee a leakage-safe comparison.

**Primary metric:** Macro F1, which weights every class equally regardless of size — the appropriate choice for a 78/12/10 imbalanced target. Balanced Accuracy, Macro Precision/Recall, and ROC-AUC were tracked as supporting evidence.

## 7.2 Baseline Cross-Validation Results

### Class-Weighting Strategy

| Model | Accuracy | Balanced Acc. | F1 Macro | ROC-AUC |
|---|---|---|---|---|
| XGBoost | 91.58% | 81.88% | 84.38% | 96.79% |
| Random Forest | 90.58% | 76.66% | 81.63% | — |
| SVM | 87.35% | 88.17% | 80.71% | — |
| Logistic Regression | 86.69% | 88.73% | 80.11% | — |
| Decision Tree | 86.05% | 74.56% | 75.02% | — |

### SMOTE Strategy

| Model | Accuracy | Balanced Acc. | F1 Macro |
|---|---|---|---|
| XGBoost + SMOTE | 91.35% | 82.92% | 84.28% |
| Random Forest + SMOTE | 90.74% | 82.62% | 83.36% |
| SVM + SMOTE | 88.14% | 86.95% | 81.35% |
| Logistic Regression + SMOTE | 87.51% | 88.65% | 81.00% |
| Decision Tree + SMOTE | 85.13% | 76.29% | 74.54% |

XGBoost led under **both** strategies with near-identical Macro F1 (84.38% vs. 84.28%), and was carried forward — alongside Random Forest, its closest competitor — into hyperparameter tuning.

## 7.3 Class Weight vs. SMOTE Comparison

The two imbalance-handling strategies were compared directly across all five candidate models on cross-validated Macro F1.


*Figure 7.3 — Cross-Validated Macro F1: Class Weight vs. SMOTE*

**Observation:** the two strategies perform comparably across every model, with SMOTE holding a slight edge for Random Forest, SVM, and Logistic Regression, while class weighting is marginally stronger for XGBoost and Decision Tree. Given the near-parity in performance, class weighting was preferred going forward — it avoids generating synthetic samples and is simpler to reproduce consistently at inference time.

## 7.4 Hyperparameter Tuning

The strongest candidates from the baseline comparison — XGBoost (both strategies) and Random Forest — were tuned using **RandomizedSearchCV** with Stratified 5-Fold cross-validation, optimising for Macro F1. The test set remained untouched throughout this stage.

| Candidate | Best CV Macro F1 |
|---|---|
| XGBoost — Class Weight (selected) | 85.33% |
| XGBoost — SMOTE | 85.05% |
| Random Forest — Tuned | 83.26% |

### Selected Configuration — XGBoost (Class Weight)

| Hyperparameter | Tuned Value |
|---|---|
| n_estimators | 500 |
| max_depth | 3 |
| learning_rate | 0.05 |
| subsample | 1.0 |
| colsample_bytree | 1.0 |
| min_child_weight | 1 |

**SELECTED — XGBoost (Class Weight, tuned) selected as the final model — highest cross-validated Macro F1 among all candidates tested.**

# 8. Final Model Evaluation

The tuned XGBoost model was evaluated once, on the 2,000-record held-out test set that had not been used during training, tuning, or cross-validation — giving a realistic estimate of production performance.

## 8.1 Held-Out Test Set Performance

| Metric | Score |
|---|---|
| Accuracy | 93.50% |
| Balanced Accuracy | 87.43% |
| F1 Macro | 88.50% |
| F1 Weighted | 93.42% |
| Precision Macro | 89.65% |
| Recall Macro | 87.43% |
| ROC-AUC (One-vs-Rest) | 98.29% |
| PR-AUC (Macro Avg., One-vs-Rest) | 94.88% |

The test Macro F1 (88.50%) exceeds the cross-validated average (85.33%), and overall accuracy (93.50%) comfortably clears the 78% no-skill baseline of always predicting the majority class — strong evidence of genuine generalisation rather than overfitting to the training folds.

## 8.2 Confusion Matrix

The confusion matrix breaks down prediction outcomes by true class to reveal exactly where the model succeeds and where it confuses classes.

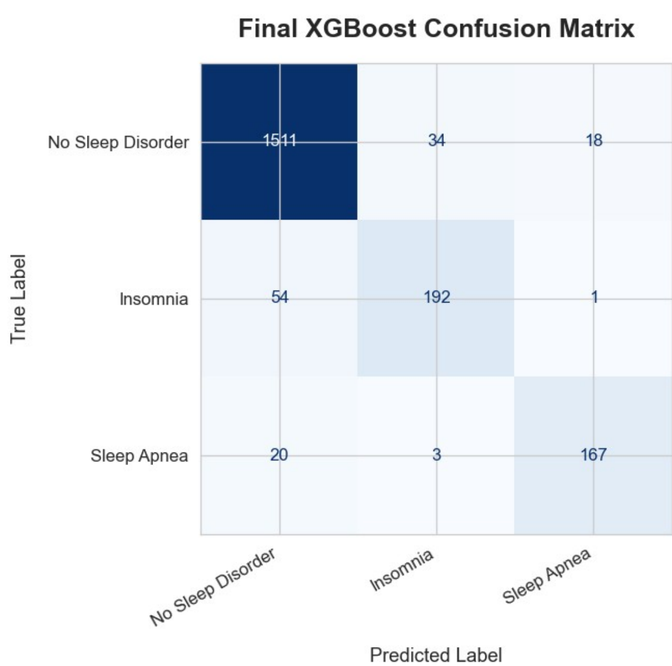


*Figure 8.2 — Final XGBoost Confusion Matrix (Test Set)*

**Key Insights:** the model correctly classifies 1,511 of 1,563 No Sleep Disorder cases and 167 of 190 Sleep Apnea cases. Insomnia is the hardest class to isolate — 54 of 247 true Insomnia cases are misclassified as No Sleep Disorder, while confusion between Insomnia and Sleep Apnea is minimal (at most 3 records each direction), showing the model rarely confuses the two disorder types with each other — only with the healthy baseline.

## 8.3 Multiclass ROC-AUC

One-vs-Rest ROC curves were plotted for each class to evaluate ranking quality independent of the classification threshold.

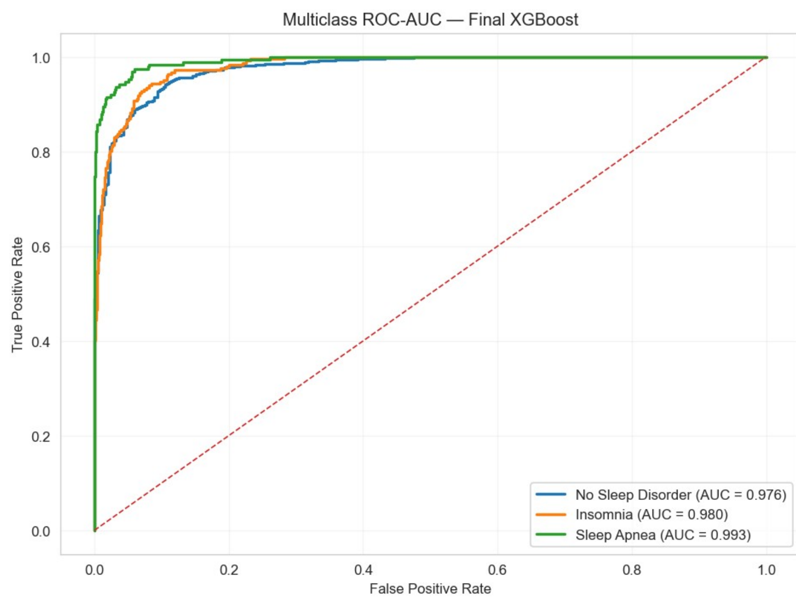


*Figure 8.3 — Multiclass ROC-AUC Curve, Final XGBoost Model*

**Insights:** all three classes achieve AUC above 0.97 — No Sleep Disorder 0.976, Insomnia 0.980, and Sleep Apnea 0.993 — indicating excellent discrimination across every decision threshold, not just the default 50% cut-off used for the confusion matrix above.

## 8.4 Precision–Recall Curves & PR-AUC

Because the target is imbalanced, Precision–Recall analysis was used alongside ROC-AUC to give a second, threshold-independent view of class-specific performance. One-vs-Rest PR curves were plotted for each class using the final XGBoost model, with particular attention to the more difficult, less-separated classes.

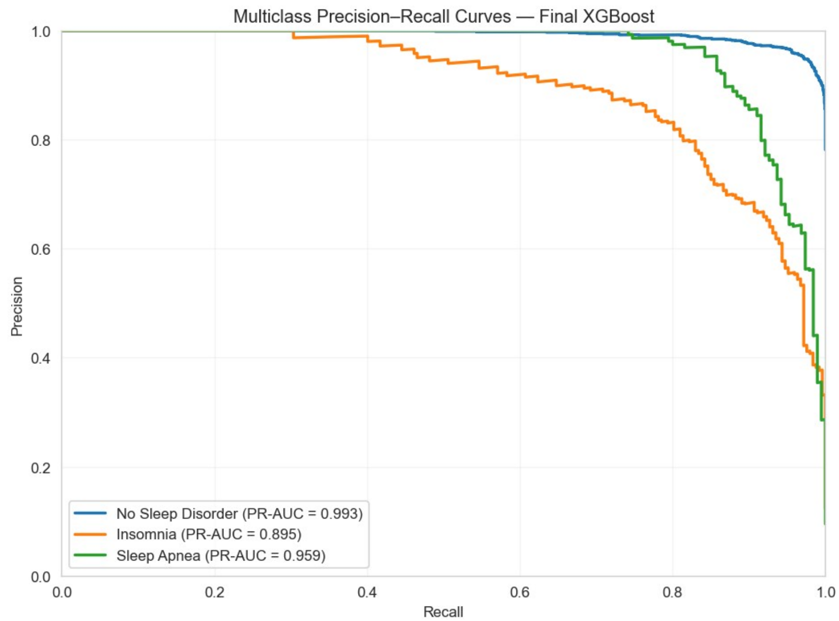


*Figure 8.4 — Multiclass Precision–Recall Curves, Final XGBoost Model*

**Insights:** PR-AUC is 0.993 for No Sleep Disorder and 0.959 for Sleep Apnea — both comfortably above 0.95 — while Insomnia trails at 0.895. Because PR-AUC is more sensitive than ROC-AUC to performance on the minority classes, this gap reinforces the confusion-matrix finding that Insomnia is the model's hardest class to separate from a healthy profile, even though its ROC-AUC alone (0.980) looked strong.

## 8.5 Class-wise Performance

Precision, Recall, and F1-score were computed independently for each class to identify which disorder is hardest for the model to detect.

| Class | Precision | Recall | F1-Score |
|---|---|---|---|
| No Sleep Disorder | 95.33% | 96.67% | 96.00% |
| Sleep Apnea | 89.78% | 87.89% | 88.83% |
| Insomnia | 83.84% | 77.73% | 80.67% |

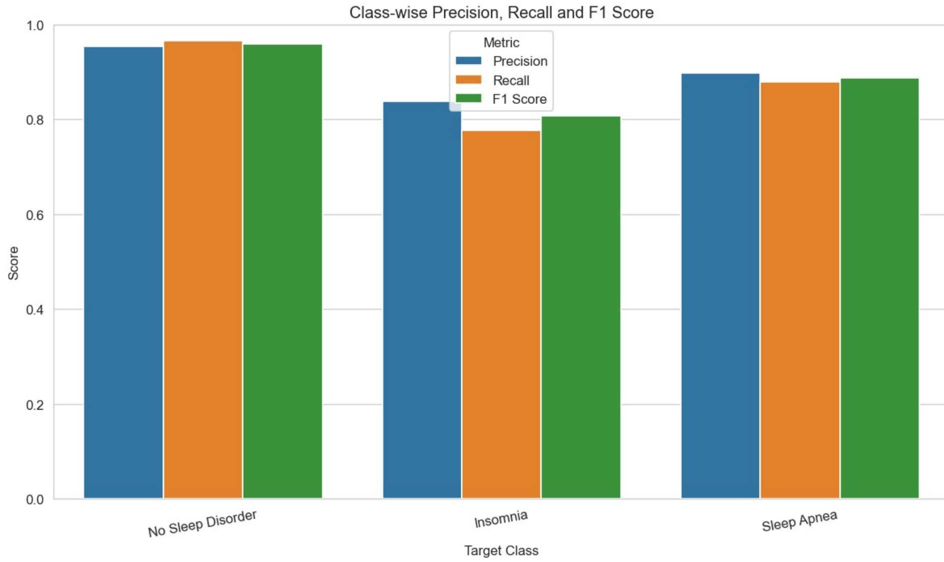


*Figure 8.5 — Class-wise Precision, Recall, and F1-Score*

**Reading the results:** No Sleep Disorder and Sleep Apnea are both detected reliably (F1 above 88%), while Insomnia remains the hardest class, most often confused with a healthy profile — consistent with the EDA finding that Insomnia's physiological signals overlap more with No Sleep Disorder than Sleep Apnea's more distinctive cardiovascular profile does.

# 9. Probability Calibration Analysis

Beyond raw classification accuracy, the reliability of the final XGBoost model's predicted probabilities was examined as a separate post-training validation experiment. Probability calibration evaluates whether the predicted probabilities correspond to the observed frequency of each sleep-disorder class — a well-calibrated classifier assigns probabilities that reflect the actual likelihood of the corresponding outcome. Calibration is evaluated separately for No Sleep Disorder, Insomnia, and Sleep Apnea using a one-vs-rest approach, combining calibration curves, quantitative probability metrics, and a comparison against sigmoid-calibrated (Platt Scaling) and isotonic-calibrated versions of the model.

## 9.1 Probability Calibration Curves

Calibration curves compare the mean predicted probability within ten uniform probability bins against the observed frequency of each class, evaluated separately for No Sleep Disorder, Insomnia, and Sleep Apnea using a one-vs-rest approach. A perfectly calibrated model would fall exactly on the diagonal reference line.

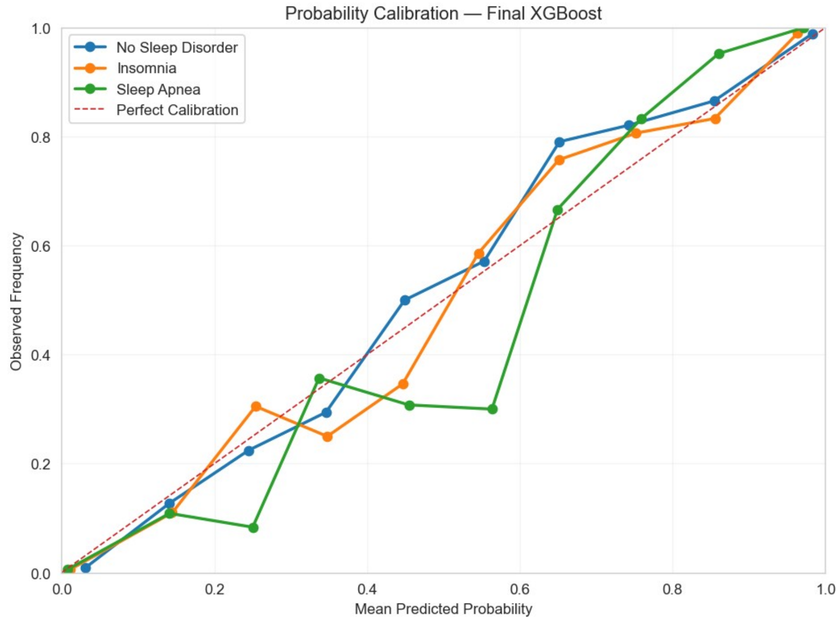


*Figure 9.1 — Probability Calibration Curves, Final XGBoost Model*

The calibration curves compare the predicted probabilities with the observed class frequencies for No Sleep Disorder, Insomnia, and Sleep Apnea. The curves generally follow the perfect calibration reference line, although deviations are visible across some probability ranges, particularly for Sleep Apnea.

The results suggest that the original XGBoost probabilities provide a useful level of probability agreement, while also indicating that some calibration error remains across individual probability ranges. Quantitative probability metrics are therefore considered alongside the calibration curves.

## 9.2 Calibration Metrics

To complement the calibration curves, Log Loss and the multiclass Brier Score were calculated on the held-out test set to quantify the quality of the predicted probability distributions. Lower values indicate better probability performance.

| Metric | Score |
|---|---|
| Multiclass Log Loss | 0.1700 |
| Multiclass Brier Score | 0.0965 |

The final XGBoost model achieved a multiclass Log Loss of 0.1700 and a multiclass Brier Score of 0.0965 on the test set. Combined with the calibration curves, the results indicate that the model's predicted probabilities show a reasonable level of agreement with the observed class frequencies, although some deviations from the perfect calibration line are present, particularly for Sleep Apnea across intermediate probability ranges.

These results provide evidence that the model probabilities are reasonably informative; however, calibration performance should be interpreted alongside the reliability curves and the model's overall classification performance.

## 9.3 Platt Scaling (Sigmoid) Calibration

To determine whether probability calibration improves the reliability of the final XGBoost model, a calibrated version was trained using CalibratedClassifierCV with 5-fold cross-validation on the training set. The sigmoid (Platt scaling) method was used as a post-training calibration approach for the multiclass probability estimates.

The test set was not used to fit the calibration model — it was used only to evaluate and compare the original and sigmoid-calibrated probability estimates.

### 9.4 Platt Scaling Calibration Results

The sigmoid-calibrated XGBoost model was evaluated using Multiclass Log Loss and Multiclass Brier Score on the held-out test set. These metrics assess the quality of the calibrated probability estimates across the three sleep-disorder classes. Lower values indicate better probability performance.

| Metric | Score |
|---|---|
| Multiclass Log Loss | 0.1941 |
| Multiclass Brier Score | 0.0988 |

The sigmoid-calibrated model achieved a higher Log Loss and Brier Score than the original, uncalibrated XGBoost model — indicating that Platt Scaling did not improve the reliability of the predicted probabilities on this held-out test set.

### 9.5 Isotonic Regression Calibration

Since sigmoid (Platt scaling) calibration did not improve the probability performance of the final XGBoost model, Isotonic Regression was evaluated as an alternative, non-parametric post-training calibration approach. As with Platt scaling, a calibrated version of the model was trained using CalibratedClassifierCV with 5-fold cross-validation on the training set, this time specifying the isotonic method — a separate isotonic calibrator is fitted for each sleep-disorder class using a one-vs-rest formulation, learning a monotonic mapping between the original XGBoost probabilities and the observed class outcomes.

As with the sigmoid experiment, the test set was not used to fit the isotonic calibrators — it was used only to evaluate the resulting calibrated probability estimates against the original, uncalibrated XGBoost probabilities.

### 9.6 Isotonic Regression Calibration Results

The isotonic-calibrated XGBoost model was evaluated using Multiclass Log Loss and Multiclass Brier Score on the held-out test set. These metrics assess whether isotonic calibration improves the quality of the predicted probabilities compared with the original XGBoost estimates. Lower values indicate better probability performance.

| Metric | Score |
|---|---|
| Multiclass Log Loss | 0.1718 |
| Multiclass Brier Score | 0.0977 |

Isotonic Regression achieved a lower Log Loss and Brier Score than Platt Scaling, placing it closer to the original, uncalibrated XGBoost model — though it still did not surpass the uncalibrated model's own probability performance.

### 9.7 Calibration Method Comparison

The uncalibrated, sigmoid-calibrated, and isotonic-calibrated XGBoost models were compared using Log Loss and the multiclass Brier Score, all evaluated on the same untouched test set. Lower values indicate better probability estimation; this three-way comparison determines whether either calibration method provides a meaningful improvement in the reliability of the predicted probabilities.

| Metric | Uncalibrated | Platt Scaling | Isotonic |
|---|---|---|---|
| Multiclass Log Loss | 0.1700 | 0.1941 | 0.1718 |
| Multiclass Brier Score | 0.0965 | 0.0988 | 0.0977 |

**Calibration Comparison and Deployment Decision**

The original XGBoost model was compared against Platt Scaling (sigmoid) and Isotonic Regression on the held-out test set. The uncalibrated model achieved a multiclass Log Loss of 0.1700 and a Brier Score of 0.0965; Platt Scaling achieved 0.1941 and 0.0988; and Isotonic Regression achieved 0.1718 and 0.0977 — higher error than the uncalibrated model on both metrics, though closer to it than Platt Scaling. Since lower values indicate better probability performance, neither calibration method improved the original XGBoost probability estimates, and the original, uncalibrated XGBoost model is retained for deployment. The Platt Scaling and Isotonic Regression experiments are documented as post-training validation analyses, while the production prediction pipeline remains unchanged.

# 10. Model Explainability (XAI)

To ensure the final XGBoost model is not a 'black box', SHAP (SHapley Additive exPlanations) was applied via **TreeExplainer** on the transformed test set to quantify how much each feature contributes to every prediction, both globally and by feature interaction.

## 10.1 SHAP Global Feature Importance

Aggregating SHAP values across all test observations, split by target class, produces a global importance ranking for every feature in the model.

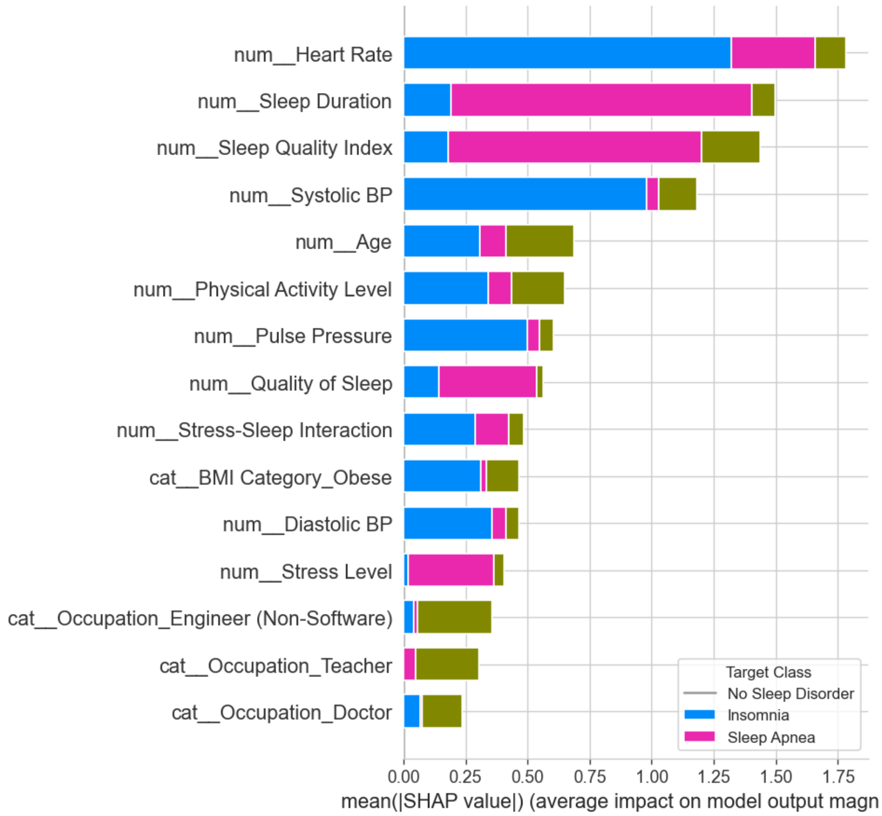


*Figure 9.1 — SHAP Global Feature Importance by Target Class*

**Insights: Heart Rate** is the single strongest predictor across all classes, followed by **Sleep Duration** and the engineered **Sleep Quality Index**. **Systolic BP** and **Age** contribute substantially, while **Physical Activity Level** and **Pulse Pressure** provide meaningful supporting signal. Demographic and occupational one-hot features (e.g. Occupation_Doctor, Occupation_Teacher) rank lowest overall — the model relies primarily on physiological and sleep-related signals rather than demographic proxies.

## 10.2 SHAP Feature Interaction Analysis

An interaction summary was generated for the model's top continuous predictors to examine whether pairs of features jointly influence predictions beyond their individual, additive effects.

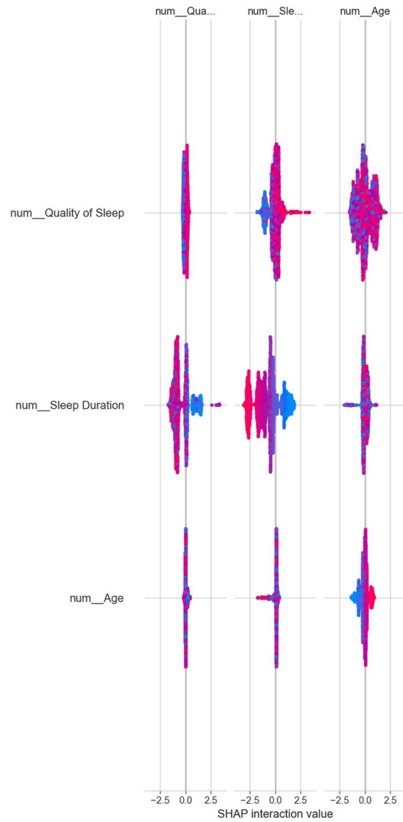


*Figure 9.2 — SHAP Feature Interaction Summary (Quality of Sleep, Sleep Duration, Age)*

**Key Insight:** Sleep Duration shows the strongest interaction effect with Quality of Sleep — the impact of one depends visibly on the level of the other, reflecting their real-world relationship as two facets of the same underlying sleep-health construct rather than fully independent signals.

### 10.3 Explainability Summary

> **Key Insight**
>
> **Heart Rate, Sleep Duration,** and the **Sleep Quality Index** dominate the model's predictions across both the global SHAP ranking and the interaction analysis. Cardiovascular indicators (**Systolic BP**, **Pulse Pressure**) provide strong secondary signal, particularly for Sleep Apnea, while demographic and occupational features contribute comparatively little. This consistent, human-readable explanation is exactly what makes the model suitable for a real-world screening context, where 'why' matters as much as 'what'.

## 11. Conclusion & Key Takeaways

### 11.1 Summary of Findings

- **XGBoost** achieved the strongest and most consistent performance among all five algorithms tested, under both class-weighting and SMOTE imbalance strategies.
- **Class weighting and SMOTE performed comparably** across every model — class weighting was marginally better for the final model and simpler to deploy, since it requires no synthetic-sample generation at inference time.

- **Heart Rate, Sleep Duration, and Sleep Quality Index** are the strongest predictors of sleep-disorder status, confirmed independently by both coefficient-level and SHAP-based explainability.
- **Sleep Apnea** is driven primarily by elevated blood pressure and heart rate; **Insomnia** by lower sleep quality and duration combined with higher stress.
- **Insomnia** remains the hardest class to separate from a healthy sleep profile — the clearest target for future feature or data improvements.

## 11.2 Final Recommendation

The tuned **XGBoost (Class Weight)** model is recommended as the deployment candidate for this classification task, combining the best held-out test performance (93.50% accuracy, 88.50% Macro F1, 98.29% ROC-AUC) with full interpretability via SHAP — a combination essential for any model intended to support health-related screening decisions.

## 11.3 Business Impact & Recommendations

- Position the model as a **screening / triage aid**, not a diagnostic system — flagged cases should be routed to a qualified healthcare provider for confirmation.
- Prioritise **Insomnia recall** in future iterations, given it is the class most often confused with a healthy profile and therefore the highest-risk false negative.
- Use the SHAP-based explanations to give end users (clinicians or individuals) a transparent, feature-level rationale alongside every prediction, not just a class label.
- Validate the pipeline on real clinical data before any production use, since the current dataset is synthetically generated.

## 11.4 Skills & Techniques Demonstrated

- Data cleaning, validation, and exploratory data analysis (pandas, seaborn, matplotlib).
- Imbalanced-classification methodology: stratified splitting, class weighting, and SMOTE, compared empirically rather than assumed.
- Feature engineering for health and lifestyle data: pulse pressure, a composite sleep-quality index, and a stress–sleep interaction term.
- Leakage-safe pipeline design using scikit-learn's ColumnTransformer and Pipeline, with resampling refit inside every cross-validation fold.
- Model development across linear (Logistic Regression), tree-based (Decision Tree, Random Forest), kernel-based (SVM), and gradient-boosted (XGBoost) paradigms.
- Hyperparameter optimisation using RandomizedSearchCV with Stratified K-Fold cross-validation.
- Explainable AI (XAI) using SHAP TreeExplainer — global feature importance and feature-interaction analysis.
- Clear, structured technical communication — translating model internals into business-relevant conclusions for non-technical stakeholders.

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*— End of Report —*